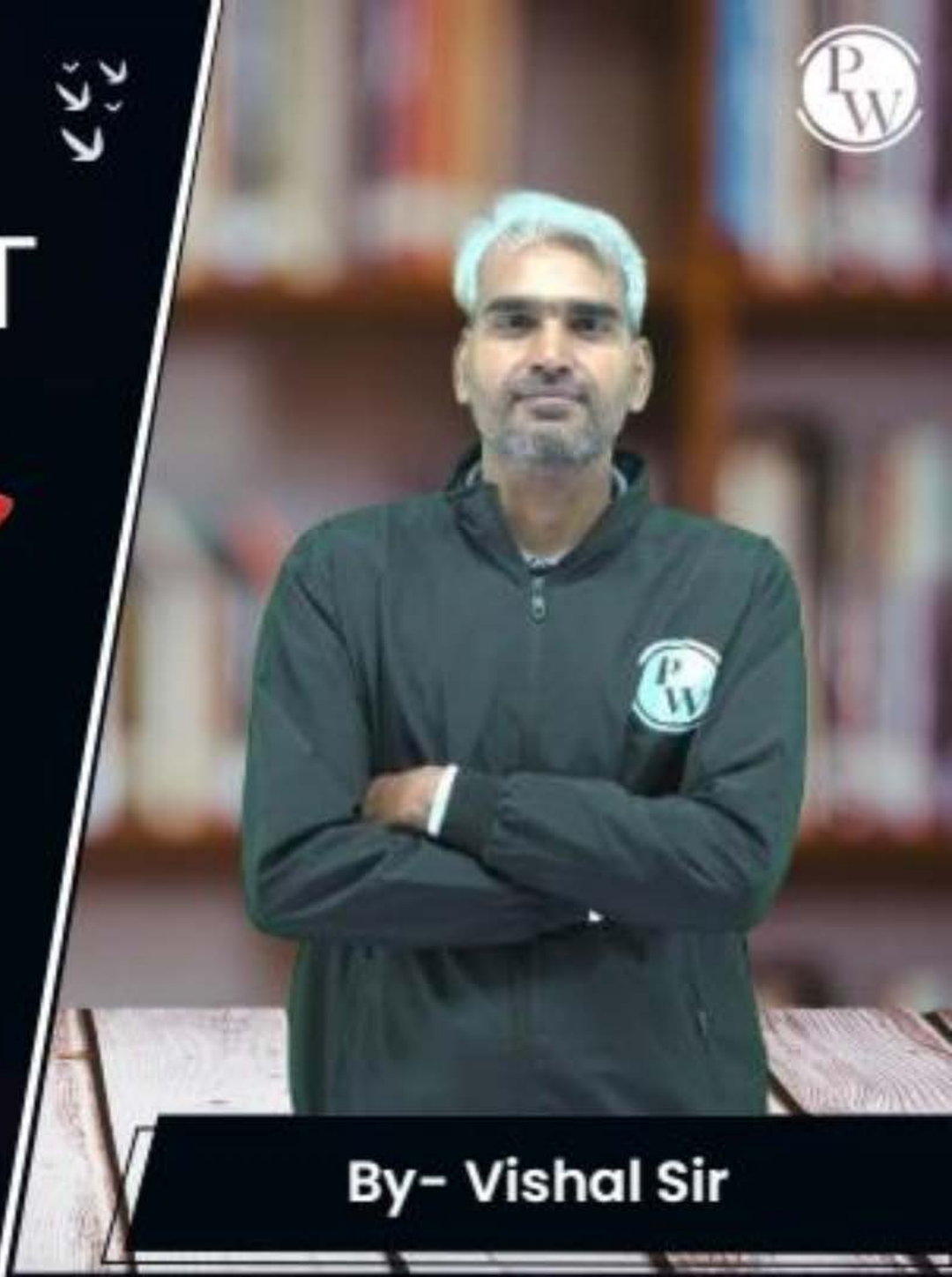
Computer Science & IT



Database Management System

Relational Model & Normal Forms

Lecture No. 13



By- Vishal Sir

Recap of Previous Lecture



Topic Second normal form (2NF)

Topic Third normal form (3NF)

Topic Boyce Codd normal form (BCNF)

Topic Identifying the normal form of the relation

Topic Decomposition of relation

Slide

Topics to be Covered

Topic Decomposition of relation



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#e.g. Given R(ABCDEF) and F={A→BCDEF, BC→ADEF, D→E , B→F}

Find the highest normal form of the relation, and if relation is not already in BCNF then decompose the relation up to BCNF.

{ 1NF }

R(A B C D E F)

$A \xrightarrow[S.K]{} BCDEF$ {BCNF}

$BC \xrightarrow[S.K]{} ADEF$ {BCNF}

$D \xrightarrow[NPA]{} E_{NPA}$ {2 NF}

$B \longrightarrow F_{NPA}$ {1 NF}

P.S.C.K

C.K. = (A), (BC)

Prime Attributes = {A, B, C}

Non-Prime Attributes = {D, E, F}

2NF + lossless Join decomposition + Dep. Preserving decomposition

R(ABCDEF)

Not in 2NF bcoz of $B \to F$

∴ Decompose w.r.t $B \to F$ $(BF)^+ = \{B, F\}$

$R_1(BF)$

$F_1 = \{B \to F_{BCNF}\}$

C.K = (B)

for lossless Join

$R_2(ABCDE)$

F_2: $A \to BCDE$ BCNF; $BC \to ADE$ BCNF; $D \to E$ 2NF (NPA → NPA)

C.K = {A, BC}

$R_2(ABCDE)$ is not in 3NF because of $D \to E$

∴ Decompose w.r.t $D \to E$ $(DE)^+ = \{D, E\}$

$R_1(BF)$

$F_1 = \{B \to F\}$ BCNF

C.K : B

$R_3(DE)$

$F_3 = \{D \to E\}$ BCNF

C.K : D

for lossless

$R_4(ABCD)$

F_4: $A \to BCD$ BCNF; $BC \to AD$ BCNF

C.K = {A, BC}

lossless Join decomposition + Dep preserving + 3NF + BCNF

#e.g. Given R(ABCD) and F={AB→C, BC→D}

Find the highest normal form of the relation, and if relation is not already in BCNF then decompose the relation up to BCNF.

$R(A\ B\ C\ D)$

$$F = \begin{cases} AB \xrightarrow{\text{S.K}} C \ \{BCNF\} \\ BC \longrightarrow D \end{cases}$$

C.K = (AB)

Prime Attributes = {A, B}

Non-prime Att = {C, D}

W.r.t

$BC \longrightarrow D$

BC → Not a Proper Subset of C.K

∴ Partial dependency can not exists

↓

Hence

$BC \longrightarrow D$

is at least in 2NF

Check for 3NF

$BC \longrightarrow D$

Neither BC is a S.K

D is a Prime Attribute

∴ transitive dependency exists.

↓

∴ Not in 3NF

$R(ABCD)$

$AB \longrightarrow C$ {BCNF}
S.K

$BC \longrightarrow D$ { it is in 2NF but not in 3NF } $\Rightarrow$ i.e 2NF

$C_K := (AB)$

$R(ABCD)$

↓

Not in 3NF bcoz of $BC \rightarrow D$

↓

∴ Decompose w.r.t. $BC \rightarrow D$

$(BCD)^+ = \{B, C, D\}$

$R_1(BCD)$

$F_1 = \{ BC \rightarrow D$ BCNF $\}$
S.K

$C_K = (BC)$

$R_2(A\,BC)$ for lossless

$F_2 = \{ AB \rightarrow C$ BCNF $\}$
S.K

$C_K = AB$

lossless Join
+
Dep. preserving
+
3NF + BCNF

#e.g. Given R(ABCDEFGHIJ) and F={AB→C, A→DE, B→F, F→GH, D→IJ}

Find the highest normal form of the relation, and if relation is not already in BCNF then decompose the relation up to BCNF.

R(A B C D E F G H I J)

$F_1 = \{$
- $AB \rightarrow C$ {BCNF} (SK)
- $A \rightarrow DE$ {1NF} (PSCK → NPA)
- $B \rightarrow F$ {1NF} (PSCK → NPA)
- $F \rightarrow GH$ (2NF) (NPA → NPA)
- $D \rightarrow IJ$ {2NF} (NPA → NPA)

$\}$

Two FDs w.r.t. 1NF. We can start decomposing w.r.t. any one of these two.

P.A : {A, B}

Ck = (AB) NPA = {C, D, E, F, G, H, I, J}

Lossless + Dep. preserving + 2NF

Lossless Dep. preserving 1NF

R(A B C D E F G H I J)

$(ADE)^+ = \{A, D, E, I, J\}$ Not in 2NF because of $A \rightarrow DE$ & $B \rightarrow F$

let us first decompose w.r.t. $A \rightarrow DE$ $(ADE)^+ = \{A, D, E, I, J\}$

$R_1(A D E I J)$
- $F_1 = \{ A \rightarrow DE$ (BCNF) (SK)
- $D \rightarrow IJ$ (2NF) (NPA → NPA)

$\}$

Ck: A

2 NF

$R_2(A B C F G H)$ — for lossless

1NF

- $AB \rightarrow C$ (BCNF) (SK)
- $B \rightarrow F$ (1NF) (PSCK → NPA)
- $F \rightarrow GH$ (2NF) (NPA → NPA)

Ck: (AB)

B → F is still in 1NF

∴ Decompose w.r.t. B → F

$(BF)^+ = \{B, F, G, H\}$

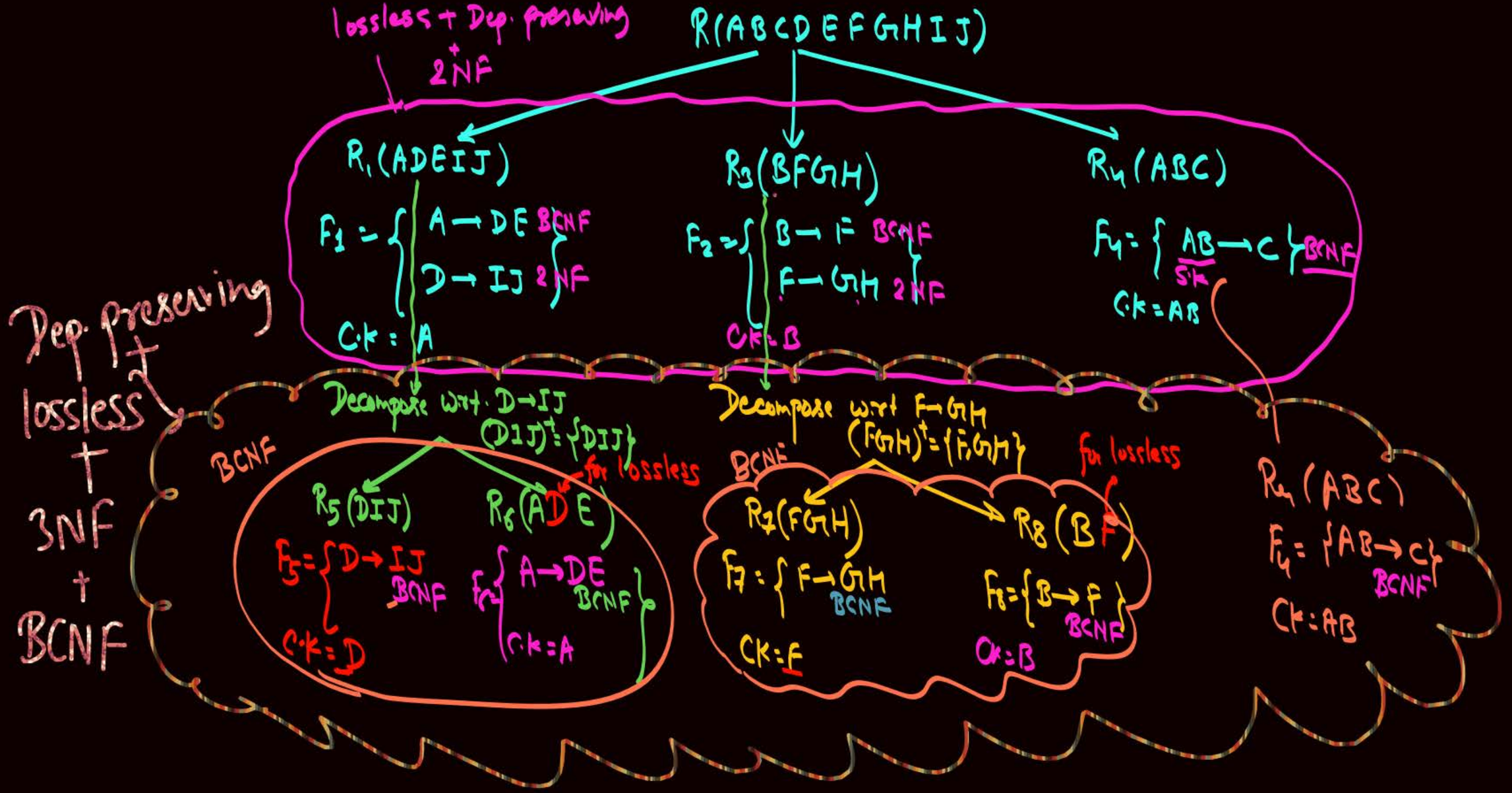
$(BF)^+ = \{B, F, G, H\}$ — for lossless

$R_3(B F G H)$
- $B \rightarrow F$ BCNF (PSCK → NPA)
- $F \rightarrow GH$ 2NF (NPA → NPA)

Ck: B

$R_4(A B C)$
- $AB \rightarrow C$ BCNF (SK)

Ck: AB


lossless + Dep. preserving
+
2NF
R(ABCDEFGHIJ)
R1(ADEIJ)
F1 = { A → DE BCNF, D → IJ 2NF }
CK : A
R3(BFGH)
F2 = { B → F BCNF, F → GH 2NF }
CK = B
R4(ABC)
F4 = { AB → C } BCNF
SK
CK = AB
Dep. preserving
+
lossless
+
3NF
+
BCNF
Decompose w.r.t. D → IJ
(DIJ)+ = {DIJ}
for lossless
BCNF
R5(DIJ)
F5 = { D → IJ BCNF }
CK = D
R6(ADE)
F6 = { A → DE BCNF }
CK = A
Decompose w.r.t F → GH
(FGH)+ = {F,GH}
for lossless
BCNF
R7(FGH)
F7 = { F → GH BCNF }
CK : F
R8(BF)
F8 = { B → F BCNF }
CK : B
R4(ABC)
F4 = {AB → C} BCNF
CK : AB

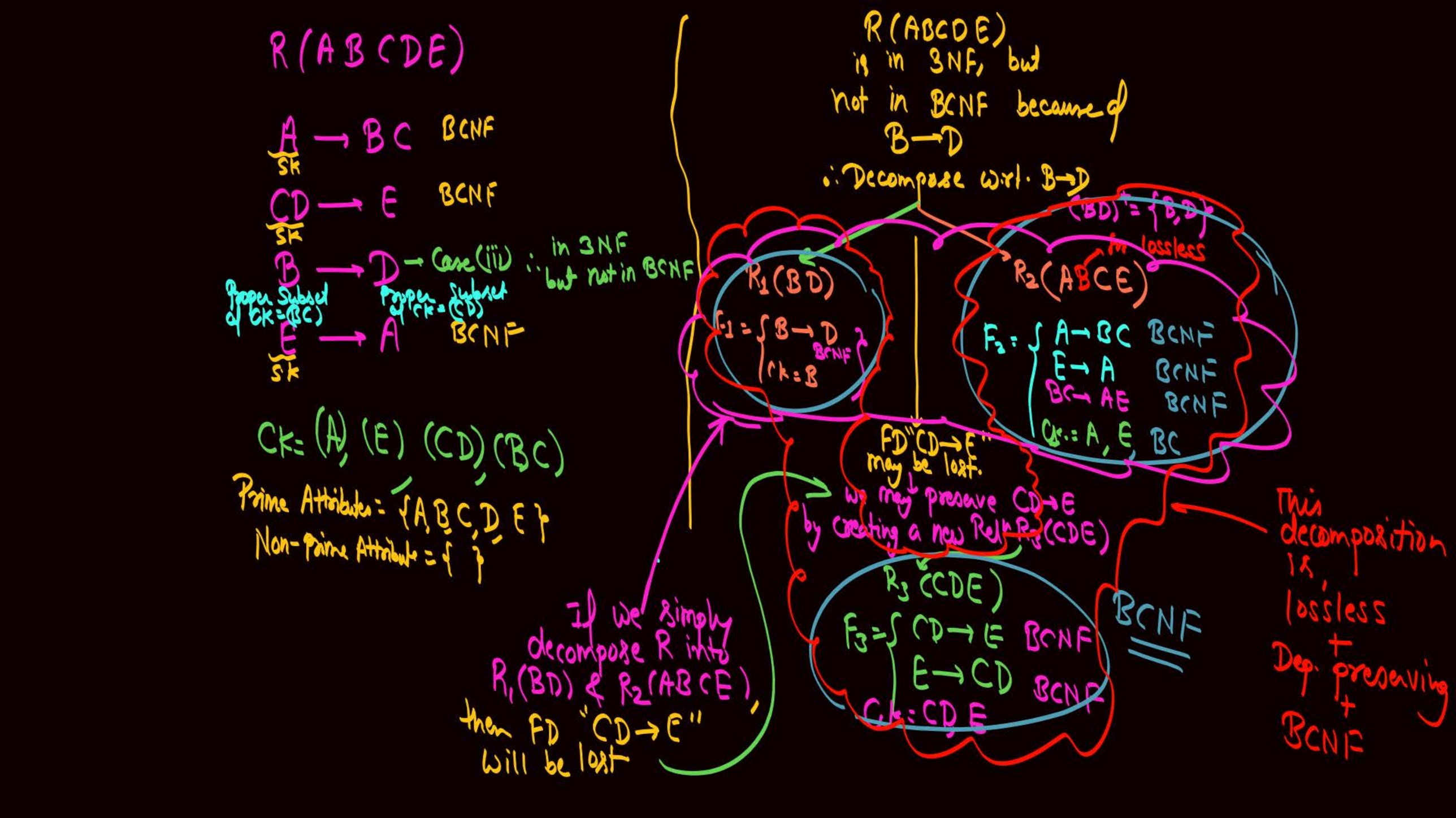
H.W.

#e.g. Given R(ABDLPT) and F={B→PT, T→L, A→D}

Find the highest normal form of the relation, and if relation is not already in BCNF then decompose the relation up to BCNF.

#e.g. Given R(ABCDE) and F:{A→BC, CD→E, B→D, E→A}

Find the highest normal form of the relation, and if relation is not already in BCNF then decompose the relation up to BCNF.


R(ABCDE)
A → BC BCNF
SK
CD → E BCNF
SK
B → D → Case (iii) : in 3NF but not in BCNF
E → A BCNF
Proper Subset of CK = (BC)
Proper Subset of CK = (CD)
SK
CK = (A), (E), (CD), (BC)
Prime Attributes = {A, B, C, D, E}
Non-Prime Attributes = { }
R(ABCDE) is in 3NF, but not in BCNF because of B → D
∴ Decompose w.r.t. B → D
R1(BD)
F1 = { B → D BCNF }
CK = B
(BD) = {B, D}
lossless
R2(ABCE)
F2 = { A → BC BCNF
E → A BCNF
BC → AE BCNF }
CK :: A, E, BC
FD "CD → E" may be lost.
we may preserve CD → E by creating a new Rel. R3(CDE)
R3(CDE)
F3 = { CD → E BCNF
E → CD BCNF }
CK : CD, E
BCNF
If we simply decompose R into R1(BD) & R2(ABCE), then FD "CD → E" will be lost
This decomposition is, lossless + Dep. preserving + BCNF

Note: ① In a relation if all attributes are prime attributes then, that relation is at least in 3NF

{Because, No partial dependency or transitive dependency is possible without non-prime attributes}



2 mins Summary

Topic Decomposition of relation

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THANK - YOU